Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main()
{
  int t;
  scanf("%d",&t);
  while(t--)
  {
      int n;
      scanf("%d",&n);
      int A[n];
      for(int i=0;i<n;i++)
      {
          scanf("%d",&A[i]);
      }
      int k;
      scanf("%d",&k);
      int c=0;
      for(int i=0;i<n;i++)
      {
          for(int j=i+1;j<n;j++)
          {
              if(A[i]-A[j]==k||A[j]-A[i]==k)
              {
                  c=1;
                  break;
              }
          }
          if(c)
          break;
      }
      printf("%d\n",c);
  }
  return 0;
}
```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Explanation

Test Case 0: N = 1

Sam buys 1 chocolate on day 1, giving us a total of 1 chocolate. Thus, we print 1 on a new line.

Test Case 1: N = 2

Sam buys 1 chocolate on day 1 and 0 on day 2. This gives us a total of 1 chocolate. Thus, we print 1 on a new line.

Test Case 2: N = 3

Sam buys 1 chocolate on day 1, 0 on day 2, and 3 on day 3. This gives us a total of 4 chocolates. Thus, we print 4 on a new line.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int t;
    scanf("%d",&t);
    while(t--)
    {
        int n,c=0;
        scanf("%d",&n);
        for(int i=0;i<=n;i++)
        {
            if(i%2!=0)
            c=c+i;
        }
        printf("%d\n",c);
    }
    return 0;
}
```

	Input	Expected	Got	
✓	3 1 2 3	1 1 4	1 1 4	✓
✓	10 71 100 86 54 40 9 77 9 13 98	1296 2500 1849 729 400 25 1521 25 49 2401	1296 2500 1849 729 400 25 1521 25 49 2401	✓

Passed all tests! ✓

We are given, n = 5, nums = [2, 10, 5, 4, 8], m = 4, and maxes = [3, 1, 7, 8].

1. For maxes[0] = 3, we have 1 element in nums (nums[0] = 2) that is ≤ maxes[0].
2. For maxes[1] = 1, there are 0 elements in nums that are ≤ maxes[1].
3. For maxes[2] = 7, we have 3 elements in nums (nums[0] = 2, nums[2] = 5, and nums[3] = 4) that are ≤ maxes[2].
4. For maxes[3] = 8, we have 4 elements in nums (nums[0] = 2, nums[2] = 5, nums[3] = 4, and nums[4] = 8) that are ≤ maxes[3].

Thus, the function returns the array [1, 0, 3, 4] as the answer.

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int s1,s2;
    scanf("%d",&s1);
    int t1[s1];
    for(int i=0;i<s1;i++)
    {
        scanf("%d",&t1[i]);
    }
        scanf("%d",&s2);
        int t2[s2];
        for(int i=0;i<s2;i++)
        scanf("%d",&t2[i]);
        for(int i=0;i<s2;i++)
        {
            int ans=0;
            for(int j=0;j<s1;j++)
        {
            if(t2[i]>=t1[j])ans++;
        }
        printf("%d\n",ans);
        }
        return 0;
}
```

	Input	Expected	Got	
✓	4 1 4 2 4 2 3 5	2 4	2 4	✓
✓	5 2 10 5 4 8 4 3 1 7 8	1 0 3 4	1 0 3 4	✓

Passed all tests! ✓